Exercise 3: SQL Fundamentals

1, SQL Joins

```
SELECT EmployeeID,
       firstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	firstName	LastName	Department	Salary	Pro-ID	Pro Name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	ongoing
3	Bob	Johnson	Finance	75000	104	financial Analysis	90000	ongoing
5	Emma	Wilson	Sales	65000	105	market Expansion	60000	Completed
6	Michael	Clerk	finance	80000	106	Risk Management	80000	Pending

2. SELECT EmployeeID,
first_name,
Last_name,
Department,
Salary,
ProjectID,
ProjectName,
Budget,
Status

FROM Employees AS A
LEFT JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;

EmployeID	firstName	lastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100 000	Completed
1	John	Doe	IT	70000	103	Cybersecurity audit	75 000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial analysis	90 000	Ongoing
4	David	Brown	IT	72000	null	null	null	null
5	Emma	Wilson	Sales	65000	105	market Expansion	65 000	Completed
6	Michael	Clark	Finance	80000	106	risk management	80 000	Pending

(3)- SELECT Project ID,
Project Name,
Budget,
Status,
EmployeeID,
FirstName,
LastName,
Department,
Salary
FROM Employees AS A
RIGHT JOIN Projects AS B
~~ON A.Employees = B.Pro~~
ON A.EmployeeID = B.EmployeeID;

Project ID	Project Name	Budget	Status	EmployeeID	FirstName	LastName	Department	Salary
101	AI Development	100000	Completed	1	John	Doe	IT	70 000
102	Employee training	50000	Ongoing	2	Alice	Smith	HR	60 000
103	cybersecurity Audit	75 000	Pending	1	John	Doe	IT	70 000
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson	Finance	75 000
105	Market Expansion	65000	Completed	5	Emma	Wilson	Sales	65 000
106	Risk Management	80000	Pending	6	Michael	Clark	Finance	80 000

⑥ SELECT EmployeeID,
FirstName,
LastName,
Department,
Salary,
ProjectID,
ProjectName,
Budget,
Status
FROM Employees AS A
FULL OUTER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID

EmployeeID	First Name	Last Name	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Completed
1	John	Doe	IT	70 000	103	cybersecurity Audit	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72 000	Null	null	null	null
5	Emma	Wilson	Sales	65 000	105	Market Expansion	65 000	Completed
6	Michael	Clark	Finance	80 000	106	Risk Management	80 000	Pending

⑦ ~~Employees~~
SELECT EmployeeID,
FirstName,
LastName,
Department,
Salary
FROM Employees
WHERE Salary > 70 000;

EmployeeID	First Name	Last Name	Department	Salary
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

⑧ SELECT EmployeeID,
First Name,
Last Name,
Department,
Salary.
FROM Employees
WHERE Department = 'IT' OR Department = 'finance';

EmployeeID	First Name	Last Name	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

⑨ SELECT ProjectID,
Project Name,
Budget,
Status
FROM Projects
WHERE Status != 'completed';

ProjectID	Project Name	Budget	Status
102	Employee Training	50 000	Ongoing
103	Cybersecurity Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	HR Management	80 000	Pending

(10) SELECT ProjectID,
Project Name,
Budget,
Status
FROM Projects
WHERE Budget > 70000 AND Status != 'completed';

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75 000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80 000	Pending

(11) SELECT EmployeeID,
firstName,
LastName,
Department,
Salary,
City
FROM Employees
WHERE City = 'New York' OR City = 'Toronto'
ORDER BY Salary DESC;

EmployeeID	firstName	LastName	Department	Salary	City
6	Mianaei	Clark	Finance	80 000	New York
1	John	Doe	IT	70 000	New York
3	Bob	Johnson	Finance	75000	Toronto

(12) SELECT EmployeeID,
first Name,
LastName,
Department,
Salary
FROM Employees
ORDER BY Salary DESC | LIMIT 3;

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	finance	80 000
3	Bob	Johnson	finance	75000
4	David	Brown	IT	72000

(13) SELECT Department,
SUM(salary) AS Total_salary
FROM Employees
GROUP BY Department
ORDER BY Total_salary DESC;

Department	Total_salary
finance	155 000
IT	142 000
HR	60 000
Sales	65000

(14) SELECT city,
AVG(Salary) AS Avg_salary
FROM Employees
GROUP BY City
HAVING AVG(salary) > 65 000;

City	Avg_Salary
New York	75 000
Toronto	75000
London	72000

(15)
```
SELECT Department,
       COUNT(EmployeeID) AS number_of_employees
FROM Employees
GROUP BY Department
HAVING COUNT(EmployeeID) > 1;
```

Department	number_of_employees
IT	2
Finance	2

(16)
```
SELECT status,
       COUNT(ProjectID) AS ProjectCount
FROM Projects
GROUP BY status
HAVING COUNT(ProjectID) >= 2;
```

Status	ProjectCount
Ongoing	2
Pending	2

(17)
```
SELECT EmployeeID,
       FirstName,
       LastName,
       SUM(Budget) AS total_budget_for_project
FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID
GROUP BY EmployeeID,
         FirstName,
         LastName
HAVING SUM(Budget) > 150 000;
```

EmployeeID	FirstName	LastName	total_budget_for_project
1	John	Doe	175000